

Weili Xu

✉ weilixu2@illinois.edu

🏠 weili-0234.github.io

🌐 github.com/weili-0234

🌐 linkedin.com/in/weili-xu-2a05662a7

306 N Wright St.

Urbana, IL, 61801

Dept. of Electrical & Computer Engineering

University of Illinois Urbana-Champaign

EDUCATION

University of Illinois Urbana-Champaign

Aug. 2023 – Present

B.S. in Computer Engineering, GPA: 3.97/4.00

Zhejiang University (Dual Degree Program)

Aug. 2023 – Present

B.S. in Computer Engineering, GPA: 4.20/4.30

Selected Courseworks (all A/A+):

ECE 408 Applied Parallel Programming, ECE 391 Computer Systems Engineering

SKILLS

Programming Language: Python, C/C++, Triton, CUDA C/C++, cuTile, Java, RISC-V Assembly

Frameworks: PyTorch, vLLM, SGLang, TensorRT-LLM, Dynamo, NIXL, MoonCake, Megatron-LM, slime

Tools: Git, Linux/UNIX, slurm, Kubernetes, Docker, GDB, PDB, Nsight Systems, Nsight Compute

EXPERIENCE

Together AI

May 2026 – Present

Research Intern, Core ML Team

San Francisco, CA

- Optimizing full stack LLM inference infrastructure for high throughput agentic traffic.

TileGym (NVIDIA's cuTile Kernel Library)

Dec. 2025 - Present

Open-Source Code Contributor

Remote

- Implemented backward-pass kernels for FlashAttention and SwiGLU in cuTile, with numerical correctness test suites and benchmarking harness; achieved >10% higher throughput than optimized Triton baseline.

University of Illinois Urbana-Champaign

Aug. 2025 – Present

Research Intern in Machine Learning Systems

Urbana, IL

ThunderAgent: A Fast, Simple, and Robust Program-Aware Agentic Inference System

- Co-first author paper, with 300+ stars on GitHub, accepted by ICML 2026 as Spotlight.
- Identified KV cache thrashing as the bottleneck in request-level LLM serving. Proposed program abstraction as the first class scheduling unit for agentic LLM inference, standardized as part of NVIDIA Dynamo 2.0.
- Achieved 1.5–3.6× serving throughput and 1.8–3.9× RL rollout throughput on agentic workloads (OpenHands, mini-SWEAgent, ToolOrchestra) on an 8×H100 node. Lifecycle-aware garbage collection yielded 4.2× disk memory savings by automatically reclaiming zombie tool resources upon program termination.

Zhejiang University

Aug. 2023 - Jul. 2025

Research Intern in Machine Learning

Hangzhou, China

AuroraLong: Bringing RNNs Back to Efficient Open-Ended Video Understanding (ICCV 2025)

- First author paper that pioneered the use of scalable Linear Attention to unlock hour-long video understanding on edge GPUs with limited VRAM, accelerated inference times by **8.13x** on RTX 3090 GPU.

Video-MMLU: A Massive Multi-Discipline Lecture Understanding Benchmark (ICCVW 2025)

- Developed a benchmark to thoroughly evaluate multimodal reasoning of **90+** video LLMs, earning Oral Presentation and Outstanding Paper Award at Knowledge-Intensive Multimodal Reasoning Workshop.

Zhejiang University-University of Illinois Urbana-Champaign Institute

Aug. 2024 – May 2025

Undergraduate Teaching Assistant

Hangzhou, China

- **ECE 220 (Computer Systems & Programming)**

Spring 2025

- **ECE 120 (Introduction to Computing)**

Fall 2024